

Personal Risk Assessment

1. Person Information Asset: Me

ID	Asset Name	Description	Sys ID	CIA Asset Value	Priority	Threat Categorization	Vulnerability Description	ARO	Controls in Place	Uncertainty	Risk Value	Controls Needed	Action Plan ID
1.	Tiwa	Person: Me	High	\$850,000	Vital	Confidential: High	Re-used passwords	High-.15	Use unique complex pass-phrase .65	Med -.7	\$75,862	Enforce MFA on all accounts; periodic credential audits	AP -01
	Tiwa	Person: Me	High	\$850,000	Vital	Confidential: High	Sharing personal details online	Med -.1	Private page to limit visibility .65	Med -.7	\$50,575	Annual privacy audit; reduce digital footprint	AP -02
	Tiwa	Person: Me	High	\$850,000	Vital	Confidential: High	Phishing and social engineering	Low -.05	Regular phishing training .6	Med -.7	\$28,900	Frequent phishing exercises	AP-03
	Tiwa	Person: Me	High	\$850,000	Vital	Integrity: Moderate	Installing unverified software	Low -.05	Install from official websites only .75	Low -.5	\$15,938	Blockers; Installing from verified sources only	AP-04
	Tiwa	Person: Me	High	\$850,000	Vital	Integrity: Moderate	Lack of file history or back ups	Med -.1	Regular backups .7	Med -.7	\$43,350	Automate daily/	AP-05

												weekly backups	
	Tiwa	Person: Me	High	\$850,000	Vital	Integrity: Moderate	Ill health	Very Low - .001	Young, active, regular checkups .75	Low - .5	\$319	Stay active. Keep up with regular check ups	AP-06
	Tiwa	Person: Me	High	\$850,000	Vital	Availability: Moderate	Hardware crash, power loss	Med - .1	UPS for critical devices. Backups .7	Med - .7	\$43,350	Cloud-based redundancy for critical files	AP-07
	Tiwa	Person: Me	High	\$850,000	Vital	Availability: Moderate	Forgotten password	Low - .05	Updated password recovery methods .8	Low - .5	\$12,750	Have backup recovery codes handy	AP-08

Person Information Asset: Me

For the Person information asset, I selected myself as the subject of analysis. Considering the CIA triad, I rated **Confidentiality: High**, **Integrity: Moderate**, and **Availability: Moderate**. The CIA asset value was estimated at \$850,000, which represents five years of potential lost income at \$110,000 per year (\$550,000), plus an additional \$150,000 in intellectual property value—reflecting the cost of transferring my professional knowledge to another person if I were to lose my position. I also included \$91,000 in 401(k) savings that might need to be withdrawn prematurely in the event of job loss, and an estimated \$59,000 in potential legal or defense costs.

Confidentiality

I rated **confidentiality as high** due to risks related to password reuse, social media activity, and phishing exposure. Previously, I often reused passwords across platforms, increasing the chance of credential-stuffing attacks, which led to a high Annualized Rate of

Occurrence (ARO) score of 0.15 and medium uncertainty of 0.7. Although I now follow stronger password practices, some residual risk remains.

My private Instagram account still contains personal identifiable information (PII) such as my birthday, location, and employer, which could be used in social engineering. For this, I assigned a medium ARO of 0.10 and medium uncertainty of 0.7.

Lastly, I considered phishing a continuing threat. From a prior assignment, I correctly identified 8 out of 10 phishing attempts, showing good awareness but not immunity. This earned a low ARO score of 0.05 and medium uncertainty of 0.5, acknowledging that attackers continuously evolve their methods.

Integrity

I rated **integrity as moderate** because certain personal habits could compromise data reliability. I occasionally install unverified software, which poses a risk of malware or system instability. I also lack a consistent file version history and backup routine, leaving my data vulnerable to corruption or accidental loss.

The Annualized Rate of Occurrence ARO was rated Low – 0.05, as incidents like these are uncommon but potentially damaging. The Uncertainty score was Low – 0.5, reflecting moderate confidence in existing controls such as antivirus use and cautious downloading but recognizing the absence of automated backups. Strengthening source verification and enabling routine cloud backups would enhance integrity.

Availability

I rated **availability as moderate** due to risks like hardware failure, power outages, or password lockouts that could limit data access. A real example was the Texas power grid failure in 2021, which caused several days of downtime.

The ARO was Medium – 0.10, since such disruptions can occur a few times a year. The Uncertainty was Medium – 0.7, given that existing safeguards—like a UPS, cloud backups, and password recovery options help but don't eliminate risk. My health was considered very low risk (0.001) due to regular checkups and an active lifestyle.

Overall Risk Reflection

Through this assessment, I learned how personal habits, such as password reuse, lack of backups, and reliance on convenience, can increase risk of exposure even for a single individual. Assigning a financial value to my own information assets made the concept of risk quantification more tangible. It reinforced that personal cybersecurity is not abstract; vulnerabilities like social media oversharing, weak authentication, and unverified software installation can translate into measurable financial and reputational loss.

2. Procedural Asset: Setting alarm before leaving the house

ID	Asset Name	Description	Sys ID	CIA Asset Value	Priority	Threat Categorization	Vulnerability Description	ARO	Controls in Place	Uncertainty	Risk Value	Controls Needed	Action Plan ID
2	Alarm Procedure	Routine procedure to arm home security system when leaving	High	\$10,000	Vital	Confidential: High	Forgetting to set the alarm before leaving	High – 0.15	Visual reminders and habit routines .65	Med – 0.7	\$868	Automate arming through smartphone or schedule-based trigger	AP -09
	Alarm Procedure	Routine procedure to arm home security system when leaving	High	\$10,000	Vital	Confidential: High	Sharing alarm PIN or code with non-household members	Med – 0.10	Code shared only with trusted family .7	Med – 0.7	\$510	Rotate PIN codes quarterly; restrict access by user	AP-10
	Alarm Procedure	Routine procedure to arm home security system when leaving	High	\$10,000	Vital	Confidential: High	Weak alarm PIN easily guessed	Med – 0.10	4-digit numeric PIN .6	Med – 0.7	\$680	Strengthen to alphanumeric PIN; add biometric or app-based control	AP-11

	Alarm Procedure	Routine procedure to arm home security system when leaving	High	\$10,000	Vital	Integrity: Moderate	Incorrect system configuration or false arming status	Low – 0.05	Occasional verification on panel .75	Low – 0.5	\$219	Enable mobile confirmation alerts and arming status notifications	AP-12
	Alarm Procedure	Routine procedure to arm home security system when leaving	High	\$10,000	Vital	Integrity: Moderate	Accidental deactivation or tampering with alarm settings	Low – 0.05	Alarm system requires passcode confirmation .8	Low – 0.5	\$150	Lock control panel; assign individual user codes	AP-13
	Alarm Procedure	Routine procedure to arm home security system when leaving	High	\$10,000	Vital	Availability: Moderate	Power outage or low battery preventing system arming	Med – 0.10	Backup battery installed .7	Med – 0.7	\$510	Test battery monthly; install UPS for main unit	AP-14
	Alarm Procedure	Routine procedure to arm home security system when leaving	High	\$10,000	Vital	Availability: Moderate	Mobile app or network failure prevents remote arming	Low – 0.05	Physical keypad backup available .8	Low – 0.5	\$150	Keep local arming method functional; verify internet connection weekly	AP-15
	Alarm Procedure	Routine procedure to arm home security system when leaving	High	\$10,000	Vital	Availability: Moderate	Human error—assuming someone else armed the system	Med – 0.10	Family communication before leaving .65	Med – 0.7	\$595	Implement alarm confirmation notifications for all users	AP-16

Procedural Asset: Setting the Alarm Before Leaving the House

For the procedural asset, I selected the routine of setting the home alarm system before leaving the house because it plays a vital role in securing our house and properties. Considering the CIA triad, I rated **Confidentiality: High, Integrity: Moderate, and Availability: Moderate**. The CIA asset value was estimated at **\$10,000**, representing the financial and safety consequences of failure to properly engage this procedure. This estimate includes \$7,000 in potential loss of property and personal belongings in the event of a burglary, \$2,000 in emotional and productivity impact related to dealing with insurance claims or law enforcement, and \$1,000 in potential system maintenance or repair costs after an incident.

Confidentiality

I rated **confidentiality as high** because the alarm system safeguards the home and, by extension, protects the private information and valuables inside. If the alarm is not properly set, unauthorized individuals could gain access to both physical and digital assets, such as documents, laptops, and external drives.

Key threats include forgetting to arm the system, sharing the alarm PIN, or using weak, predictable codes. These risks occur occasionally but could have significant consequences. I assigned an ARO of 0.10 (medium) because forgetting to arm the alarm or sharing the code can happen a few times per year. The Uncertainty score is 0.7 (medium), reflecting confidence in current mitigations, such as quarterly PIN rotation, smartphone confirmation alerts, and individual user codes, but acknowledging that human error cannot be fully eliminated.

Integrity

I rated **integrity as moderate**, as human or configuration errors could affect the reliability of the alarm procedure. For instance, selecting the wrong mode (such as “Stay” instead of “Away”) could leave areas unmonitored, or a visitor unfamiliar with the system could accidentally disarm it.

These errors are relatively rare, so I assigned a Low ARO of 0.05, while the Uncertainty score is 0.5 (low), since regular mobile confirmations and monthly system tests have significantly improved accuracy and confidence in system performance. Together, these measures ensure the alarm functions as intended and maintains its integrity over time.

Availability

I rated **availability as moderate** because the reliability of this procedure depends on both user consistency and technical functionality. Power outages, connectivity failures, or low system batteries can temporarily prevent the alarm from arming or transmitting status updates. In my case, a brief power surge once caused the system to reboot, leaving it offline until manually reset.

Given that these disruptions occur occasionally, I assigned a Medium ARO of 0.10. The Uncertainty score is 0.7 (medium), since safeguards like a backup battery, uninterruptible power supply (UPS), and a family confirmation routine greatly improve reliability, but the potential for unforeseen technical failures remains.

Overall Risk Reflection

Through this assessment, I learned that even a simple procedural habit can have a profound impact on overall household security. Setting the alarm before leaving the house is not merely a physical safety measure, it is also a crucial information assurance practice, protecting the confidentiality and availability of the assets within the home.

Quantifying the asset's value made me realize how procedural discipline contributes to cybersecurity at the physical layer. Neglecting such a routine can indirectly expose confidential data, financial records, and digital systems to unauthorized access. This exercise reinforced that cybersecurity extends beyond computers and networks, it includes the everyday behaviors and controls that safeguard both physical and digital environments.

Moving forward, I plan to continue refining this procedure by integrating automated arming schedules, multi-user verification, and routine system audits. These improvements not only reduce risk but also demonstrate how consistent, security-focused habits create tangible protection for personal assets and data.

3. Data Asset: Driver's License

ID	Asset Name	Description	Sys ID	CIA Asset Value	Priority	Threat Categorization	Vulnerability Description	ARO	Controls in Place	Uncertainty	Risk Value	Controls Needed	Action Plan ID
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3	Driver's License	Physical and digital identification card	High	\$150,000	Vital	Confidential: High	Loss or theft of physical ID	High – 0.15	Kept in secure wallet; limited carrying .65	Med – 0.7	\$13,388	Lockable storage; avoid carrying when unnecessary	AP-17
	Driver's License	Physical and digital identification card	High	\$150,000	Vital	Confidential: High	Exposure of ID photo or number online	Med – 0.10	Avoid posting ID documents or scans .7	Med – 0.7	\$7,650	Enable privacy settings; avoid email sharing of IDs	AP-18
	Driver's License	Physical and digital identification card	High	\$150,000	Vital	Confidential: High	Phishing or impersonation using ID details	Low – 0.05	Verify legitimacy of ID requests .6	Med – 0.7	\$5,100	Use watermark on digital copies before sharing	AP-19
	Driver's License	Physical and digital identification card	High	\$150,000	Vital	Integrity: Moderate	Forged or altered digital copy	Low – 0.05	Share only verified PDFs/scans .75	Low – 0.5	\$2,813	Add digital watermark; share via secure portals	AP-20
	Driver's License	Physical and digital identification card	High	\$150,000	Vital	Integrity: Moderate	Expired or inaccurate ID data	Med – 0.10	Renew license promptly .7	Med – 0.7	\$7,650	Create renewal reminder system	AP-21
	Driver's License	Physical and digital identification card	High	\$150,000	Vital	Availability: Moderate	Loss during travel or emergency	Low – 0.05	Keep scanned copy in	Low – 0.5	\$2,250	Maintain notarized backup in	AP-22

									secure cloud .8			secure location	
	Driver's License	Physical and digital identification card	High	\$150,000	Vital	Availability: Moderate	DMV system outage preventing renewal	Very Low – 0.00 1	Plan renewals early .85	Low – 0.5	\$34	Renew months earlier ahead of expiry	AP-23

Data Asset: Driver's License

For the data asset, I selected my **Driver's License** as the subject of analysis, because it contains a lot of personal Identifiable Information (PII). Considering the CIA triad, I rated **Confidentiality: High**, **Integrity: Moderate**, and **Availability: Moderate**. The CIA asset value was estimated at **\$150,000**, reflecting the potential financial, legal, and reputational impact of identity theft or misuse. This estimate includes the cost of restoring compromised personal identity (estimated as \$20,000), potential fraudulent financial activity if my license were used for impersonation (approximately \$100,000 in possible losses or credit damage), and additional legal or administrative expenses (around \$30,000) required to reestablish my identity and credibility.

Confidentiality

I rated **confidentiality as high** because my driver's license contains several pieces of personally identifiable information (PII), such as my full name, address, date of birth, and license number, which could be exploited for identity theft or fraud. The primary threats include physical loss or theft, unauthorized sharing of ID images, and phishing attempts that request identity verification.

Before this assessment, I occasionally uploaded ID scans to online forms without confirming site security, which created exposure to credential theft or misuse. I have since limited online sharing and ensure that document uploads occur only through encrypted or verified portals.

The ARO was rated Medium – 0.10, since accidental oversharing or phishing attempts could occur several times a year depending on online activity. The Uncertainty score was Medium – 0.7, reflecting that while existing safeguards, like private storage, and cautious sharing, are effective, human error and social engineering remain unpredictable.

Integrity

I rated **integrity as moderate**, as risks such as data tampering or the use of altered copies are possible but manageable. A forged or modified version of my license could be used to impersonate me or create legal complications. Similarly, failing to update an expired license could lead to discrepancies or service denial.

To maintain integrity, I securely store all digital copies, monitor expiration dates, and use watermarked or redacted versions when sharing documents online to prevent unauthorized duplication.

The ARO was rated Low – 0.05, as deliberate forgery or modification is relatively uncommon. The Uncertainty was Low – 0.5, since established controls like watermarking, encryption, and verification provide reliable protection, though the potential for misuse cannot be entirely ruled out.

Availability

I rated **availability as moderate**, since temporary loss of access to my driver's license (either physical or digital) could delay identity verification for travel, employment, or legal purposes. Scenarios include losing the physical card, forgetting to renew, or DMV system outages.

To reduce these risks, I keep a secure, encrypted digital backup in the cloud and set renewal reminders well in advance. These controls minimize downtime and ensure access even during disruptions.

The ARO was rated Medium – 0.10, since occasional events like misplaced items or technical outages can occur annually. The Uncertainty was Medium – 0.7, reflecting moderate confidence in the resilience of current safeguards but recognizing potential dependence on third-party systems like cloud storage or DMV availability.

Overall Risk Reflection

This assessment deepened my understanding of the sensitivity and value of personal identification assets. While replacing a license is inexpensive, the secondary costs of identity misuse which includes financial, emotional, and reputational, can be severe. Assigning tangible ARO and Uncertainty scores revealed how easily small lapses, such as reusing IDs online or ignoring renewal reminders, can compound into measurable risks.

I learned that maintaining confidentiality and integrity for identity documents requires awareness, layered controls, and consistency, not just physical safety. Simple actions like rotating document storage locations, watermarking scans, and limiting online submissions significantly reduce exposure and preserve the reliability and availability of this critical personal asset.

4. Software Asset: Gmail

ID	Asset Name	Description	Sys ID	CIA Asset Value	Priority	Threat Categorization	Vulnerability Description	ARO	Controls in Place	Uncertainty	Risk Value	Controls Needed	Action Plan ID
4	Gmail (Emails)	Email account used for personal and professional communication	High	\$600,000	Vital	Confidential: High	Weak or reused password could allow unauthorized access to mailbox	High – 0.15	Strong password and 2FA enabled .8	Med – 0.7	\$30,600	Enforce MFA on all linked accounts; enable password breach alerts	AP-24
	Gmail (Emails)	Email account used for personal and professional communication	High	\$600,000	Vital	Confidential: High	Phishing or malicious email attachments	Med – 0.10	Awareness training and spam filter .65	Med – 0.7	\$35,700	Conduct periodic phishing simulations; verify sender authenticity	AP-25
	Gmail (Emails)	Email account used for personal and professional communication	High	\$600,000	Vital	Confidential: High	Accidental sharing of sensitive information (PII, attachments)	Low – 0.05	Limit auto-complete recipients; verify before sending .6	Med – 0.7	\$20,400	Implement DLP tools; label and classify sensitive messages	AP-26

	Gmail (Emails)	Email account used for personal and professional communication	High	\$600,000	Vital	Integrity: High	Spoofed or tampered emails (integrity compromise)	Med – 0.10	Email headers and SPF/DKIM protection .6	Med – 0.7	\$40,800	Enable DMARC enforcement; monitor for suspicious sender patterns	AP-27
	Gmail (Emails)	Email account used for personal and professional communication	High	\$600,000	Vital	Integrity: High	Sync errors or corrupted messages	Low – 0.05	Regular account sync and data backup .75	Low – 0.5	\$11,250	Automate email archiving or export backups to secure cloud	AP-28
	Gmail (Emails)	Email account used for personal and professional communication	High	\$600,000	Vital	Availability: Moderate	Account lockout or password loss	Med – 0.10	Recovery email and phone verified .7	Med – 0.7	\$30,600	Store recovery codes securely offline; use password manager	AP-29
	Gmail (Emails)	Email account used for personal and professional communication	High	\$600,000	Vital	Availability: Moderate	Google service outage or connectivity issue	Low – 0.05	Backup offline email via client (e.g., Outlook/Thunderbird) .8	Low – 0.5	\$9,000	Set up local mail sync or alternative email account for redundancy	AP-30

Software Asset: Gmail (Emails)

For the communication information asset, I selected my Gmail account, which serves as the central hub for both personal and professional correspondence. Considering the CIA triad, I rated **Confidentiality: High, Integrity: High, and Availability: Moderate.**

The CIA asset value was estimated at **\$600,000**, reflecting the cumulative value of the information stored within the account and its

potential role in enabling unauthorized access to other systems. This value includes approximately \$300,000 in data sensitivity, based on exposure of personal and financial communications, \$200,000 in potential indirect losses if the account were used to compromise other linked accounts (such as bank or work systems), and \$100,000 in reputational and recovery costs from identity theft or impersonation scenarios.

Confidentiality

I rated **confidentiality as high** due to the volume and sensitivity of information stored in my Gmail account. Emails often contain personally identifiable information (PII), banking confirmations, account credentials, and password reset links to other platforms. The main threats to confidentiality include weak or reused passwords, phishing attacks, and unauthorized access through compromised devices.

Before taking this course, I occasionally reused passwords across different accounts, which increased my exposure to credential-stuffing attacks. This led me to assign a Medium ARO score of 0.10, since password reuse and phishing threats can occur several times a year. However, I have since adopted stronger password practices and implemented two-factor authentication (2FA) to reduce exposure. I also regularly review Gmail's activity logs to monitor for suspicious sign-ins or access attempts.

The Uncertainty score was rated Medium – 0.7, since while 2FA and regular monitoring are effective safeguards, human error and evolving phishing techniques continue to pose risks. Gmail's strong security infrastructure greatly mitigates the risk, but the dynamic nature of phishing and credential theft justifies maintaining vigilance.

Integrity

I rated **integrity as high**, as email content can be manipulated, spoofed, or corrupted through unauthorized access or synchronization errors. A common integrity risk involves email spoofing, where attackers impersonate trusted senders to trick recipients or damage reputations. Additionally, accidental deletions or sync errors could compromise important information.

To mitigate these issues, I rely on Gmail's built-in authentication and sender verification features such as SPF, DKIM, and DMARC, which verify legitimate senders and flag potential tampering. I also periodically back up critical messages to a secure cloud folder to preserve message integrity.

The ARO was rated Low – 0.05, since these incidents are relatively rare and Gmail provides reliable protection. The Uncertainty was rated Medium – 0.7, reflecting that although current controls are strong, not all third-party email clients or linked services maintain equivalent security standards. Therefore, while integrity risks are infrequent, maintaining consistent backups and sender verification remains essential.

Availability

I rated **availability as moderate**, primarily because continuous access to my Gmail account is essential for both personal and professional operations. A temporary loss of access—due to password lockouts, Google outages, or compromised recovery settings—could disrupt communications and delay important tasks. For example, I rely on Gmail for verification codes, school assignments, and professional correspondence, so even a short outage could affect my productivity.

To reduce this risk, I have secured my recovery email and phone number, stored backup recovery codes in a password manager, and downloaded offline copies of key messages. However, full availability still depends on Google's infrastructure and network reliability, which are external to my direct control.

The ARO was rated Medium – 0.10, since login issues or brief outages occur occasionally, though not frequently. The Uncertainty was rated Medium – 0.7, as my confidence in current controls is strong but conditional upon Google's uptime and the integrity of my recovery credentials. These measures ensure resilience but still require active monitoring to maintain consistent access.

Overall Risk Reflection

Through this assessment, I realized how email accounts serve as both gateways and repositories for personal information, making them one of the most valuable and vulnerable digital assets. Assigning a financial value to my Gmail account clarified the real-world cost of potential compromise, from identity theft and reputational loss to unauthorized access of linked systems. I learned that

cybersecurity is not limited to technical controls, it also depends on personal vigilance in recognizing phishing, maintaining strong authentication, and securing recovery mechanisms. This analysis reinforced that email security is foundational to digital identity protection, and small proactive steps, like enabling MFA, verifying senders, and archiving sensitive messages, can significantly reduce the likelihood of costly incidents.

5. Hardware Asset: External Hard Drive

ID	Asset Name	Description	Sys ID	CIA Asset Value	Priority	Threat Categorization	Vulnerability Description	ARO	Controls in Place	Uncertainty	Risk Value	Controls Needed	Action Plan ID
5	Hard Drive	External data storage for personal and professional files	High	\$2,500	Vital	Confidential: High	Loss or theft of unencrypted drive	High – 0.15	Physical storage in safe location .6	Med – 0.7	\$255.00	Enable full-drive encryption (BitLocker/VeraCrypt); use password-protected access	AP-31
	Hard Drive	External data storage for personal and professional files	High	\$2,500	Vital	Confidential: High	Improper disposal (data remnants after deletion)	Med – 0.10	Securely delete files using software tools .65	Med – 0.7	\$148.75	Use certified data wiping tools or physical destruction before disposal	AP-32
	Hard Drive	External data storage for personal and	High	\$2,500	Vital	Confidential: High	Malware-infected backups	Low – 0.05	Antivirus protection active .6	Med – 0.7	\$85.00	Scan drives before backup; isolate	AP-33

		professional files										infected media	
	Hard Drive	External data storage for personal and professional files	High	\$2,500	Vital	Integrity: High	File corruption or physical damage	Med – 0.10	Periodic integrity checks .65	Med – 0.7	\$148.75	Use SMART monitoring tools; verify backup integrity regularly	AP-34
	Hard Drive	External data storage for personal and professional files	High	\$2,500	Vital	Integrity: High	Unintentional overwriting or deletion	Low – 0.05	Manual verification before transfer .68	Low – 0.5	\$60.00	Enable file versioning or recycle-bin recovery	AP-35
	Hard Drive	External data storage for personal and professional files	High	\$2,500	Vital	Availability: Moderate	Hardware failure (mechanical or electrical)	Med – 0.10	Secondary backup drive in use .7	Med – 0.7	\$127.50	Implement 3-2-1 backup rule (local + cloud + offsite)	AP-36
	Hard Drive	External data storage for personal and professional files	High	\$2,500	Vital	Availability: Moderate	Power surge or overheating	Low – 0.05	Surge protector used .75	Low – 0.5	\$46.88	Maintain proper ventilation; use UPS during transfers	AP-37
	Hard Drive	External data storage for personal and professional files	High	\$2,500	Vital	Availability: Moderate	Interface or cable failure	Low – 0.05	Regular cable inspection .70	Low – 0.5	\$56.25	Keep spare cables; avoid excessive bending and strain	AP-38

Hardware Asset: External Hard Drive

For the hardware asset, I selected my **external hard drive** as the subject of analysis. Considering the CIA triad, I rated **Confidentiality: High, Integrity: Moderate, and Availability: Moderate**. The CIA asset value was estimated at **\$2,500**, reflecting the cost of the device itself and, more importantly, the potential financial and reputational losses associated with the data it contains. This estimate includes \$1,500 in data value, representing sensitive personal files, academic work, and professional materials that could be exposed or stolen, \$500 in replacement and recovery costs, and \$500 in potential productivity losses due to downtime or data reconstruction.

Confidentiality

I rated **confidentiality as high** because my external hard drive stores personal, academic, and professional data, including files that contain personally identifiable information (PII) and confidential project materials. The main threats to confidentiality include loss or theft of an unencrypted drive, improper disposal, and malware-infected backups.

Before this assessment, I had not encrypted my external drives, meaning that if they were lost or stolen, anyone could access the data simply by plugging them into another device. I have since implemented full-drive encryption using BitLocker, which ensures data remains protected even if the physical hardware is compromised. I also learned that deleting files does not completely erase data, recovery tools like Recuva can restore them. To address this, I now use secure file deletion methods and plan to physically destroy drives before disposal.

The ARO was rated Medium – 0.10, as incidents like misplaced or unencrypted devices can occur periodically, especially during travel or device upgrades. The Uncertainty score was Medium – 0.7, reflecting moderate confidence in encryption and disposal practices but acknowledging that human error (e.g., failing to re-encrypt or securely wipe data) can still occur.

Integrity

I rated **integrity as high** because physical media are prone to data corruption, overwriting, or accidental deletion, and infected drives can spread malware between systems. In the past, I backed up files from multiple sources without verifying authenticity, which

increased the chance of corrupted or incomplete data being stored. Additionally, I did not perform regular verification of backup integrity, which meant potential data loss could have gone unnoticed.

To mitigate these issues, I now perform routine integrity checks, maintain verified backups across both cloud and physical locations, and use checksum validation tools to confirm file authenticity. I also rely on S.M.A.R.T. monitoring to detect early hardware degradation before it leads to failure.

The ARO was rated Low – 0.05, as data corruption and integrity loss are infrequent but potentially severe when they occur. The Uncertainty was rated Medium – 0.7, because while multiple safeguards exist, physical media remain inherently vulnerable to gradual wear, power interruptions, or unnoticed bit rot over time.

Availability

I rated **availability as moderate**, given the potential for hardware failure, accidental damage, or power surges that could disrupt access to files. I once experienced a drive disconnect mid-transfer, resulting in incomplete backups; a reminder that physical drives can fail suddenly and unpredictably.

To counter this, I follow the 3-2-1 backup principle, maintaining three copies of data (original, local backup, and cloud copy) across two types of storage media, with one copy stored offsite. I also use surge protectors, maintain proper ventilation to prevent overheating, and perform routine testing of drives to ensure accessibility.

The ARO was rated Medium – 0.10, since temporary access disruptions or disconnections can occur occasionally, while the Uncertainty was Medium – 0.7, recognizing that while redundancy greatly enhances resilience, hardware degradation and external conditions can still introduce some unpredictability.

Overall Risk Reflection

Through this assessment, I learned how easily physical media can become a cybersecurity liability if not managed properly. While the hard drive itself is relatively inexpensive, the data it holds represents significant personal and professional value. This exercise reinforced the idea that data security extends beyond digital systems, it includes how physical devices are protected, maintained,

and eventually discarded.

Assigning a monetary value to the hard drive underscored that data loss, theft, or corruption can translate directly into financial and reputational damage. I also recognized that convenience-driven behaviors, such as saving all files to a single location or skipping encryption, increases vulnerability. Moving forward, I plan to maintain encrypted, redundant backups, implement regular data verification checks, and adopt secure disposal practices to protect this essential information asset.

6. Networking Device Asset: Wi-Fi Router

ID	Asset Name	Description	Sys ID	CIA Asset Value	Priority	Threat Categorization	Vulnerability Description	ARO	Controls in Place	Uncertainty	Risk Value	Controls Needed	Action Plan ID
6	Wi-Fi Router	Primary home network gateway enabling internet connectivity	High	\$5,000	Vital	Confidential: High	Default or weak admin password allows unauthorized configuration access	High – 0.15	Unique password set; admin access limited .7	Med – 0.7	\$255	Enforce complex random passphrase; disable remote admin; change annually	AP-39
	Wi-Fi Router	Primary home network gateway enabling internet connectivity	High	\$5,000	Vital	Confidential: High	Outdated firmware exposes known security flaws	Med – 0.10	Manual updates when prompted .65	Med – 0.7	\$297.50	Schedule quarterly firmware checks; enable auto-update if supported	AP-40
	Wi-Fi Router	Primary home network gateway enabling	High	\$5,000	Vital	Confidential: High	Unsecured or outdated wireless encryption (e.g., WEP)	Med – 0.10	WPA2/WPA 3 active .7	Med – 0.7	\$255	Verify WPA3 enabled; rotate Wi-Fi key and SSID yearly	AP-41

		internet connectivity											
	Wi-Fi Router	Primary home network gateway enabling internet connectivity	High	\$5,000	Vital	Confidential: High	IoT device exposure through same subnet	Med – 0.10	IoT devices password-protected .6	Med – 0.7	\$340	Segment IoT devices to guest VLAN; disable UPnP	AP-42
	Wi-Fi Router	Primary home network gateway enabling internet connectivity	High	\$5,000	Vital	Confidential: High	Logging or metadata leakage from router analytics	Low – 0.05	Limited diagnostic data shared .7	Low – 0.5	\$112.50	Disable cloud telemetry; clear router logs periodically	AP-43
	Wi-Fi Router	Primary home network gateway enabling internet connectivity	High	\$5,000	Vital	Integrity: Moderate	Unauthorized configuration changes or DNS hijacking	Med – 0.10	Admin access restricted; DNS default .6	Med – 0.7	\$340	Use trusted DNS; enable DNSSEC	AP-44
	Wi-Fi Router	Primary home network gateway enabling internet connectivity	High	\$5,000	Vital	Integrity: Moderate	Firmware tampering or installation of rogue software	Low – 0.05	Vendor-verified updates .75	Low – 0.5	\$93.75	Validate checksum before firmware install; download from official site	AP-45
	Wi-Fi Router	Primary home network gateway enabling	High	\$5,000	Vital	Integrity: Moderate	Guest network misuse or	Med – 0.10	Guest Wi-Fi limited	Med – 0.7	\$340	Enable guest isolation; review connected-	AP-46

		internet connectivity					unmonitored activity		bandwidth .6			device logs monthly	
	Wi-Fi Router	Primary home network gateway enabling internet connectivity	High	\$5,000	Vital	Availability: Moderate	Hardware failure, overheating, or dust accumulation	Med – 0.10	Proper ventilation; surge protector .7	Med – 0.7	\$255	Schedule routine cleaning; replace unit every 4 years	AP-47
	Wi-Fi Router	Primary home network gateway enabling internet connectivity	High	\$5,000	Vital	Availability: Moderate	Power outage or ISP service interruption	Low – 0.05	UPS and backup hotspot .8	Low – 0.5	\$75	Maintain cellular failover; monitor ISP uptime	AP-48
	Wi-Fi Router	Primary home network gateway enabling internet connectivity	High	\$5,000	Vital	Availability: Moderate	Denial-of-Service (DoS) or network flooding	Low – 0.05	Built-in firewall active .7	Low – 0.5	\$75	Enable DoS protection; coordinate with ISP for mitigation	AP-49
	Wi-Fi Router	Primary home network gateway enabling internet connectivity	High	\$5,000	Vital	Availability: Moderate	Misconfiguration leading to internet outage	Med – 0.10	Default configuration backed up .65	Med – 0.7	\$297.50	Export and store config backups; document network settings	AP-50

Networking Device Asset: Wi-Fi Router

For the networking infrastructure asset, I selected my **Wi-Fi router** as the subject of analysis. Considering the CIA triad, I rated **Confidentiality: High, Integrity: Moderate, and Availability: Moderate**. The CIA asset value was estimated at **\$5,000**, reflecting the combined cost of the physical device, replacement labor, and the potential financial and reputational damage that could result if the

router were compromised. This includes approximately \$2,000 in data exposure impact (due to unauthorized network access or interception), \$2,000 in productivity loss caused by downtime or reconfiguration, and \$1,000 in indirect recovery and mitigation costs, such as service restoration, professional diagnostics, or hardware replacement.

Confidentiality

I rated **confidentiality as high** because the Wi-Fi router serves as the primary gateway connecting all home and personal devices to the internet. It processes sensitive traffic such as online banking credentials, emails, and cloud data synchronization. If compromised, an attacker could intercept or monitor this traffic, capturing personal communications or login credentials.

Key threats include weak or default admin passwords, outdated firmware, and unsecured wireless encryption. Before this assessment, I rarely updated firmware or changed the router's administrative password, assuming my home network was inherently secure. I have since learned that routers are common targets precisely because users neglect patches and updates.

To mitigate these risks, I now enforce WPA3 encryption, disable remote management, and use a complex randomized passphrase for both admin and Wi-Fi access. I also renamed the SSID to conceal personal identifiers such as the router brand or last name.

The ARO was rated Medium 0.10, since exposure from weak credentials or outdated firmware can realistically occur once or twice a year if not properly maintained. The Uncertainty was rated Medium – 0.7, reflecting solid confidence in current controls but recognizing residual human error, such as delayed updates or password reuse, can still occur.

Integrity

I rated **integrity as moderate**, because configuration settings and DNS information can be tampered with to redirect legitimate traffic to malicious destinations. For instance, DNS hijacking could lead users to spoofed banking or login pages, resulting in credential theft. Similarly, unauthorized changes to firewall rules or port configurations could silently weaken security.

Previously, I did not back up my router configuration, meaning accidental resets or corruption could require hours to re-secure the network. I have since created an encrypted backup of configuration files and switched to trusted DNS providers such as Cloudflare with DNSSEC enabled to ensure data authenticity.

The ARO was rated Low – 0.05, as such tampering or corruption events are rare, especially with strong administrative controls in place. The Uncertainty was rated Medium – 0.7, since although router security has improved, DNS-level attacks and firmware vulnerabilities remain evolving risks outside the user’s full control.

Availability

I rated **availability as moderate**, since continuous Wi-Fi connectivity is essential for work, school, and home systems. Disruptions from hardware failure, power surges, overheating, or firmware corruption can cause downtime and loss of productivity. I previously experienced a full day of network loss due to a corrupted firmware update, which interrupted remote work and communication.

To mitigate these risks, I now maintain a backup mobile hotspot as a failover, connect the router to a surge protector, and place it in a well-ventilated location to prevent overheating. Regular firmware updates and reboots maintain optimal performance, and I keep a spare router available to minimize downtime.

The ARO was rated Medium – 0.10, since temporary outages or hardware issues can occur occasionally. The Uncertainty was rated Medium – 0.7, acknowledging strong reliability measures but recognizing that environmental factors or ISP-level outages remain partially beyond user control.

Overall Risk Reflection

Through this assessment, I learned that the Wi-Fi router, often overlooked as a simple household device, is, in fact, a critical cybersecurity asset. It controls the boundary between private and public networks and, if compromised, can undermine the confidentiality and integrity of all connected systems. The exercise reinforced that maintaining security requires regular firmware patching, strong credential management, and segmentation of connected devices such as IoT cameras or smart assistants, which often lack robust protection.

Assigning a monetary value to the router highlighted that even inexpensive hardware can represent substantial risk due to its role in the broader network ecosystem. I realized that securing this single device can drastically reduce exposure for all others. Moving forward, I plan to perform quarterly network audits, ensure automatic firmware updates, and rotate Wi-Fi credentials annually. This

process demonstrated that cybersecurity begins at the network's foundation, the router, and that consistent maintenance, not just configuration, is key to long-term protection.

Overall Risk Posture

Based on this comprehensive assessment across assets such as personal identity, procedures, digital systems, and physical devices, my overall risk posture can be characterized as moderately strong with targeted opportunities for improvement.

My current control environment demonstrates a clear awareness of cybersecurity principles and proactive implementation of protective measures, such as strong authentication practices, data encryption, redundant backups, and procedural safeguards for physical and digital assets. The average control effectiveness across all assets is approximately 65–75%, reflecting well-established but not yet fully optimized controls.

From a confidentiality perspective, high-value assets (like Gmail, Wi-Fi router, and hard drive) are protected through encryption, 2FA, and restricted access, though periodic patching, social engineering vigilance, and network segmentation remain areas for reinforcement. Integrity risks are mostly moderate, mitigated through verified sources, backups, and digital signatures, though configuration and data verification routines could be more automated. Availability risks are low-to-moderate, supported by UPS backups, redundant storage, and recovery planning, ensuring resilience against hardware failures or outages.

Overall, my environment aligns well with the CIA triad, confidentiality being the strongest pillar, followed by availability, then integrity. My residual risks are largely human-behavior or maintenance-based rather than structural weaknesses. With incremental improvements, such as automated firmware updates, broader network isolation, and periodic security audits, I am positioned to maintain a mature, risk-aware, and resilient cybersecurity posture that balances protection with practicality.